Linear Mixed Models (H1a, H2a, H3a)

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```r
# Load libaries
library(tidyverse)
library(dplyr)
library(lmerTest)
library(lme4)
```

```r
# Load Interleaf data
df <- read.csv("../../Data/df_interleaf_measures.csv")
```

```r
# Sum Contrasts: Recode condition order as hai = -0.5 and hh = 0.5
# Prevents the use of dummy or treatment contrasts,
# Interpretation: Slope represents raw difference between conditions
# Each intercept is the grand mean of the dv in the condition
```

```r
df$shared_condition_order <- as.factor(as.character(df$shared_condition_order))
contrasts(df$shared_condition_order) <- c(-0.5, 0.5)
```

```r
# Contrast Matrix
contrasts(df$shared_condition_order)
```

```
##             [,1]
## with_ai    -0.5
## without_ai  0.5
```

```r
# Check that everything is there and working
attributes(df$shared_condition_order)
```

```
## $levels
## [1] "with_ai"    "without_ai"
##
## $class
## [1] "factor"
##
## $contrasts
##             [,1]
## with_ai    -0.5
## without_ai  0.5
```

```r
# H1: A Flow model Difference between human-human and human-AI condition
flow_model <- lmer(flow_rating ~ shared_condition_order + chat_number +
                      shared_prompt_order +
                    (1 + shared_condition_order | participant_ID) +
                    (1 | dyad_ID),
                    data = df)
summary(flow_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: flow_rating ~ shared_condition_order + chat_number + shared_prompt_order +
##     (1 + shared_condition_order | participant_ID) + (1 | dyad_ID)
##    Data: df
##
## REML criterion at convergence: 2978.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.88082 -0.51951  0.05313  0.58316  2.31970
##
## Random effects:
##  Groups         Name                    Variance Std.Dev. Corr
##  participant_ID (Intercept)             243.03   15.589
##                 shared_condition_order1 460.03   21.448   -0.42
##  dyad_ID        (Intercept)              51.33    7.165
##  Residual                               275.24   16.590
## Number of obs: 336, groups:  participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##                         Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)              38.5510     3.9864  61.9851   9.671 5.37e-14 ***
## shared_condition_order1  -3.6103     3.7723  40.8231  -0.957    0.344
## chat_number               2.7887     0.4155 264.9901   6.712 1.16e-10 ***
## shared_prompt_order       1.2956     0.8140 250.0091   1.592    0.113
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) shr__1 cht_nm
## shrd_cndt_1 -0.227
## chat_number -0.416  0.008
## shrd_prmpt_ -0.461 -0.001 -0.105
```

```r
# H2: A Flow model Difference between human-human and human-AI condition
social_presence_model <- lmer(social_presence_rating ~ shared_condition_order +
                                 chat_number + shared_prompt_order +
                    (1 + shared_condition_order | participant_ID) +
                    (1 | dyad_ID),
                    data = df)
summary(social_presence_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: social_presence_rating ~ shared_condition_order + chat_number +
##     shared_prompt_order + (1 + shared_condition_order | participant_ID) +      (1 | dyad_ID)
```

```
##    Data: df
##
## REML criterion at convergence: 3015.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2307 -0.4239  0.1043  0.5440  2.9684
##
## Random effects:
##  Groups         Name                    Variance Std.Dev. Corr
##  participant_ID (Intercept)             238.32   15.438
##                 shared_condition_order1 622.33   24.947   -0.60
##  dyad_ID        (Intercept)              16.63    4.078
##  Residual                               318.55   17.848
## Number of obs: 336, groups:  participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##                          Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)               53.5712     3.9151 105.5686  13.683  < 2e-16 ***
## shared_condition_order1  -2.0537     4.3140  40.9850  -0.476    0.637
## chat_number                1.9150     0.4467 265.4293   4.287 2.54e-05 ***
## shared_prompt_order        1.1817     0.8757 250.1912   1.349    0.178
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) shr__1 cht_nm
## shrd_cndt_1 -0.331
## chat_number -0.455  0.007
## shrd_prmpt_ -0.506 -0.001 -0.104
```

```
# H3: Affect Difference between human-human and human-AI condition
panas_model <- lmer(panas_rating ~ shared_condition_order + chat_number +
                      shared_prompt_order +
                    (1 + shared_condition_order | participant_ID) +
                    (1 | dyad_ID),
                    data = df)
summary(panas_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: panas_rating ~ shared_condition_order + chat_number + shared_prompt_order +
##     (1 + shared_condition_order | participant_ID) + (1 | dyad_ID)
##    Data: df
##
## REML criterion at convergence: 2733.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5278 -0.4693  0.0278  0.5234  2.8305
##
## Random effects:
##  Groups         Name                    Variance Std.Dev. Corr
##  participant_ID (Intercept)             153.29   12.381
##                 shared_condition_order1  77.80    8.821   -0.68
```

```
##  dyad_ID        (Intercept)              78.19    8.842
##  Residual                               138.87   11.784
## Number of obs: 336, groups:  participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##                         Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)              57.3499     3.3452  47.4706  17.144  < 2e-16 ***
## shared_condition_order1   0.3548     1.8725  40.7504   0.189  0.85065
## chat_number               0.8839     0.2900 278.2036   3.048  0.00252 **
## shared_prompt_order       0.7869     0.5781 250.0983   1.361  0.17468
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) shr__1 cht_nm
## shrd_cndt_1 -0.288
## chat_number -0.346  0.011
## shrd_prmpt_ -0.392 -0.001 -0.103
```